Rain-Based Traffic Alert System

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Milestone 1: System Architecture Design & Simulation

1. Introduction

Road safety becomes critical during rainfall due to reduced visibility, slippery roads, and sudden changes in driving conditions. Continuous monitoring of weather and traffic conditions is essential to prevent accidents and ensure safe driving.

2. Actual Problem

Traditional systems lack real-time detection and response to rain-related hazards. Drivers often continue normal behavior despite risky conditions, leading to accidents, while authorities have limited visibility to enforce timely safety measures.

3. Solution Expected

A smart system using rain, traffic density, and proximity sensors with time-stamped data is needed to detect hazardous conditions. The system should classify risk levels, trigger alerts, and enable real-time actions through dashboards and smart infrastructure to improve road safety during rainfall.

4. Proposed Solution

The presented system is an intelligent traffic monitoring system that is based on an IoT and is aimed at increasing road safety in the rainy season. It is a system that incorporates various sensors to keep a real time check on the environmental and traffic conditions.

The parameters that are gathered by the system include:

1. Rain sensor (modeled by potentiometer) intensity of the rainfall.
2. The presence of traffic with the help of a digital input sensor (push button).
3. Ultrasonic sensor to determine the proximity of vehicle.
4. DHT 22 sensor of environmental temperature.
5. The sensor data obtained is sent to a microcontroller (ESP32) and processed and analyzed. According to a pre-defined logic, the system categorizes road conditions in four levels of risks:
6. Safe, Caution, High Risk, Critical.

These categories aid in determining the hazardous driving circumstances. The processed information is then forwarded to a cloud platform where it is used to visualize the input and alerts can be sent to the drivers and traffic authorities. This allows preventive safety and enhances the general traffic control in case of bad weather.

5. System Components

5.1 Sensors Used

1. Rain Sensor (Potentiometer):
2. The intensity of rainfall is simulated by giving analog values. The greater the values, the greater the rainfall.
3. Traffic Sensor (Push Button):
4. Works as an online input to show the availability of vehicles or not.
5. Ultrasonic Sensor (HC-SR04):
6. Determines the space between cars, which assists in identifying the possible crash cases.
7. Temperature Sensor (DHT22):
8. Monitors environmental temperature that may affect the road conditions (e.g. slippery road).

5.2 Microcontroller

The central processing unit is applied to ESP32. It gathers information on all sensors then calculates it in real-time and implements decision logic to calculate risk levels.
It is also involved in WiFi communication and cloud integration.

5.3 Cloud Platform

ThingSpeak is applicable in:

- Real-time data storage
- Graphical representation of data.
- Monitoring historical trends

It will enable them to see the difference in rainfall, traffic, distance, and temperature with time.

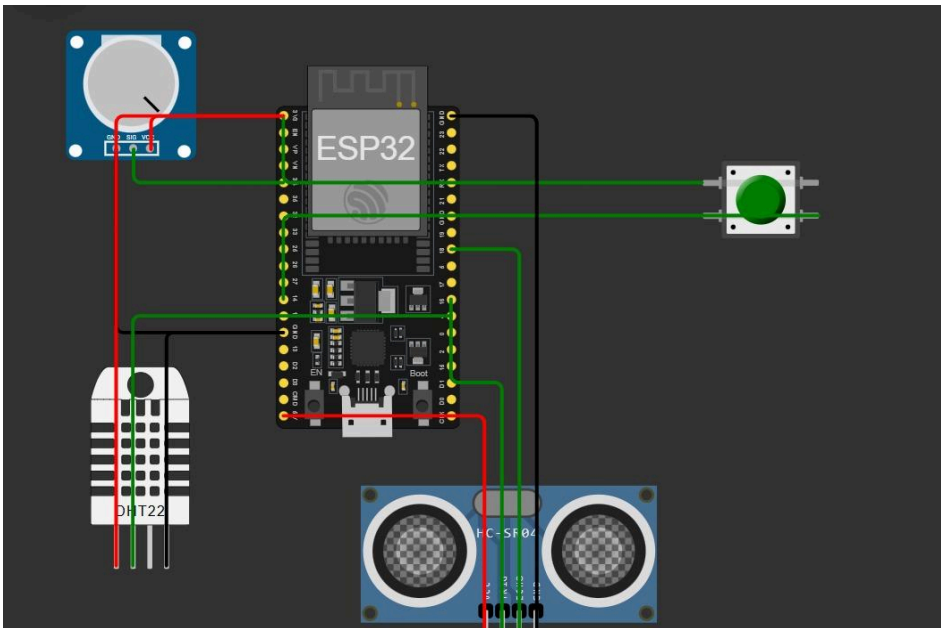
5.4 Communication

The ESP32 is connected to ThingSpeak via a sensor reading sent to it through its inbuilt WiFi module.

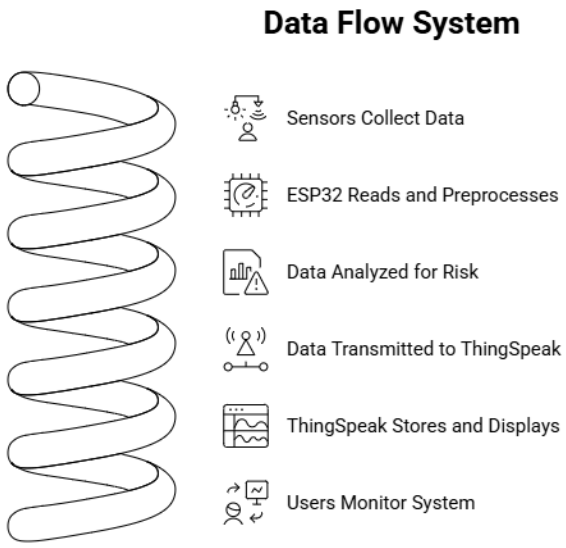
Periodic transmissions of data with the help of the HTTP requests allow almost real-time monitoring.

6. System Architecture Diagram

6.1 Circuit Diagram



7. Data Flow



8. Code

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <DHT.h>
#define WIFI_SSID "Wokwi-GUEST"
#define WIFI_PASS ""
#define API_KEY "MY307PGVNJ7D0UF8"
#define SERVER "http://api.thingspeak.com/update"
#define RAIN_PIN 34
#define TRAFFIC_PIN 14
#define TRIG 16
#define ECHO 18
#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

int rainValue = 0;
int traffic = 0;
long distance = 0;
float temperature = 0;

String rainStatus, trafficStatus, distanceStatus, risk;
unsigned long lastUploadTime = 0;
```

```
void setup() {
  Serial.begin(115200);

  pinMode(TRAFFIC_PIN, INPUT_PULLDOWN);
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  dht.begin();

  Serial.println("Connecting to WiFi...");
  WiFi.begin(WIFI_SSID, WIFI_PASS);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nWiFi Connected!");
}

long getDistance() {
  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG, LOW);

  long duration = pulseIn(ECHO, HIGH);
  return duration * 0.034 / 2;
}

void loop() {

  rainValue = analogRead(RAIN_PIN);
  traffic = digitalRead(TRAFFIC_PIN);
  distance = getDistance();
  temperature = dht.readTemperature();
```

```
if (rainValue < 1000) {
  rainStatus = "LOW";
} else if (rainValue < 2500) {
  rainStatus = "MEDIUM";
} else {
  rainStatus = "HIGH";
}

if (traffic == 1) {
  trafficStatus = "TRAFFIC PRESENT";
} else {
  trafficStatus = "NO TRAFFIC";
}

if (distance > 50) {
  distanceStatus = "SAFE";
} else if (distance > 20) {
  distanceStatus = "MEDIUM";
} else {
  distanceStatus = "DANGEROUS";
}

if (rainStatus == "HIGH" && distanceStatus == "DANGEROUS" && traffic == 1) {
  risk = "CRITICAL";
}
else if (rainStatus == "MEDIUM" || distanceStatus == "MEDIUM") {
  risk = "HIGH";
}
else if (rainStatus == "LOW") {
  risk = "CAUTION";
}
else {
  risk = "SAFE";
}


Serial.println("------ DATA ------");
Serial.println("Rain: " + String(rainValue) + " (" + rainStatus + ")");
Serial.println("Traffic: " + trafficStatus);
Serial.println("Distance: " + String(distance) + " cm (" + distanceStatus + ")");
```

```
  Serial.println("Temperature: " + String(temperature));
  Serial.println("Risk Level: " + risk);
  Serial.println("------------------");


  if (millis() - lastUploadTime >= 20000) {

    if (WiFi.status() == WL_CONNECTED) {

      HTTPClient http;

      String url = String(SERVER) + "?api_key=" + API_KEY +
                   "&field1=" + String(rainValue) +
                   "&field2=" + String(traffic) +
                   "&field3=" + String(distance) +
                   "&field4=" + String(temperature);

      http.begin(url);
      int httpResponseCode = http.GET();

      if (httpResponseCode > 0) {
        Serial.println("Uploaded to ThingSpeak ");
      } else {
        Serial.println("Upload failed ");
      }

      http.end();
    }

    lastUploadTime = millis();
  }

  delay(1000);
}
```

9. Simulation Setup

The implementation of the system was done using:

Wokwi IoT Simulator

Components Simulated:

- ESP32 microcontroller
- Potentiometer (Rain simulation)
- Push Button (Traffic detection)
- The device is known as Ultrasonic Sensor (Distance measurement).
- DHT22 (Temperature sensing)

The simulation environment is a replica of real world behavior of sensors, which allows testing without hardware.

10. Working Logic

Rain Classification

- 0 - 1000 : Low Rainfall
- 1000 - 2500 : Medium Rainfall
- 2500 : High Rainfall

Traffic Detection

Button Pressed then Traffic Present
If Button is not pressed then no Traffic

Distance Classification

- 50 cm : Safe Distance
- 20 - 50 cm : Moderate Distance
- <20 cm : Dangerous Distance

Risk Calculation Logic

The system is a combination of various conditions to identify general risk:

- Large rainfall + hazardous distance + traffic = CRITICAL.
- Moderate distance or medium rainfall rate : HIGH RISK.
- Low rain : CAUTION
- Otherwise : SAFE

This is a rule-based method which models real-world safety systems of decision-making.

11. Output

```
Connecting to WiFi...
.......
WiFi Connected!
------ DATA ------
Rain: 4095 (HIGH)
Traffic: NO TRAFFIC
Distance: 158 cm (SAFE)
Temperature: -1.30
Risk Level: SAFE
------------------
------ DATA ------
Rain: 4095 (HIGH)
Traffic: NO TRAFFIC
Distance: 158 cm (SAFE)
Temperature: -1.30
Risk Level: SAFE
------------------
------ DATA ------
Rain: 4095 (HIGH)
Traffic: TRAFFIC PRESENT
Distance: 159 cm (SAFE)
Temperature: -1.30
Risk Level: SAFE
------------------
------ DATA ------
Rain: 4095 (HIGH)
Traffic: NO TRAFFIC
Distance: 158 cm (SAFE)
Temperature: -1.30
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```
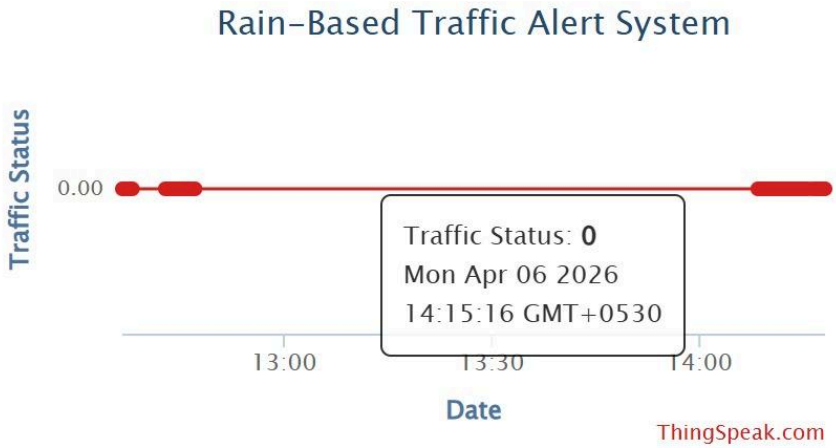
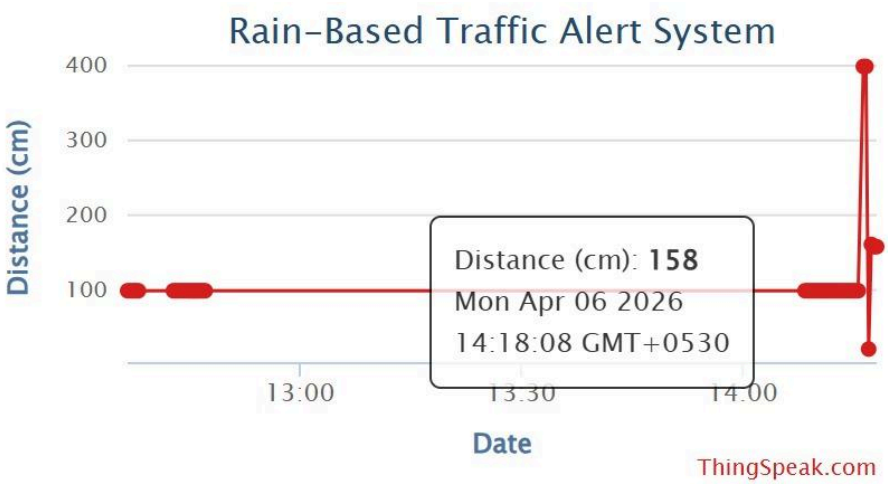
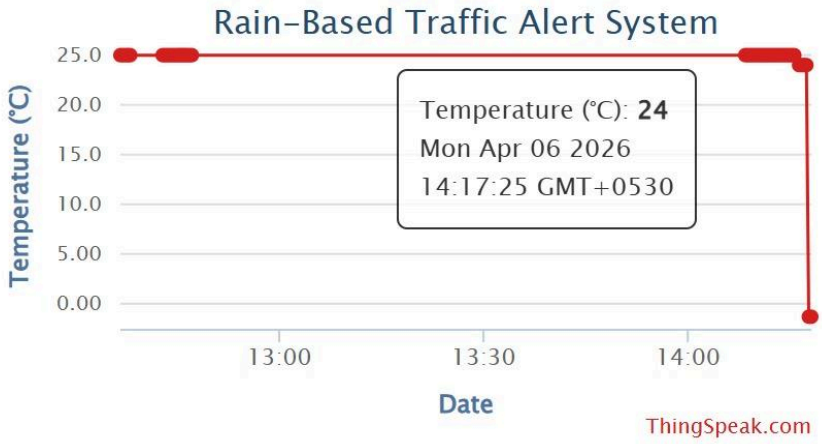

12. ThingSpeak Output

Fields Used:
Field 1 : Rain Intensity
Field 2 : Traffic Status
Field 3 : Distance
Field 4 : Temperature
The data will be presented as graphs, so users can:

- Monitor trends in the course of time.
- Analyze environmental changes
- Check the behaviour of the monitors remotely.







13. Conclusion

The proposed system effectively illustrates a real-time traffic alert system on the basis of IoT through simulation. It combines various sensors, a microcontroller, and a cloud platform to detect the road conditions and categorize the level of risk successfully. The system offers a scalable platform on which smart traffic management can be done and it can be extended by: The integration of machine learning predictive analysis models, Traffic detection with real-time cameras, Implementation in intelligent cities. On the whole, the project demonstrates how an IoT can enhance road safety and minimize the risks of accidents in unfavorable weather.